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An in-depth analysis of code smell agglomeration stability

Maintaining and evolving code

- Changing the code is a challenging activity:
  - Understand the code and its complexity
  - Class dependencies and ripple effects

Code Smell

- Symptoms of developer’s decisions that may lead to code quality degradation:
  - Complexity;
  - Cohesion;
  - Coupling;
  - Modularity;
  - Size;
  - Errors and Faults.

Code Smell Agglomerations

- When two or more code smell occurs on the same piece of code. In our study, in a class.
- Heterogeneous
  - Two or more smells of different types
- Homogeneous
  - Two or more smells of the same type
- Isolated
  - Only one smell
- Clean
  - No smell

Related Works

- Palomba et al. (2017) investigated which agglomeration is more frequent in the source code, and later (2018) investigated how they are spread along the systems;
- Santana et al. (2024) investigated how agglomerations impact source code metrics;
- Khomh et al. (2012) investigated how anti-patterns impacts change-proneness and fault-proneness.

Goal: Provide Evidences of Code Smell Agglomeration Stability

Research Question

- RQ3: What specific combination of code smells change more frequently and with more intensity?

Methodology summary

Mined two-year of commits of 30 open-source Java systems from GitHub between 2021 and 2023;

9 code smells;

Parsed the diff files to identify:

If commit created or deleted class;

Number of lines of code added, deleted and non-functional changes (cosmetic);

Mann-Whittney U + Holm-Bonferroni Correction + Cliff’s Delta

Preliminary Results

Commits and committers

- Clean classes usually receives more commits than smelly classes;
- For specific smell combinations, we could not reject the null hypothesis, indicating that there is no difference between the number of commits and the number of different committers.

#Add, #Del, Churn and Cosmetic

- For the Heterogeneous agglomeration, we have found that:

Agglomeration Type	Modification Type	Agglomeration Pair	HB	Delta
Heterogeneous	#Add	DicoIc-LcDico	0.0001	-0.467
		DicoIc-DicoLm	0.0002	-0.455
	Churn	DicoIc-IcLm	0.001	-0.564
	Cosmetic	DicoIc-IcLm	0.001	-0.572

- Meanwhile, for Isolated Classes, we found that Data Classes usually are more stable than other isolated smelly classes.

Threats to Validity

- Detection of smells in only one system release;
- Two year mining timespan;
- Systems selected;
- Effect of system size.

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Future Works

- Further explore other measurements of stability;
- Explore the impact of agglomerations on coupled classes;
- Explore changes in terms of smelly methods;
- Explore how Generative AI impacts on the maintanance and evolution of agglomerations.

Thank you! :)

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